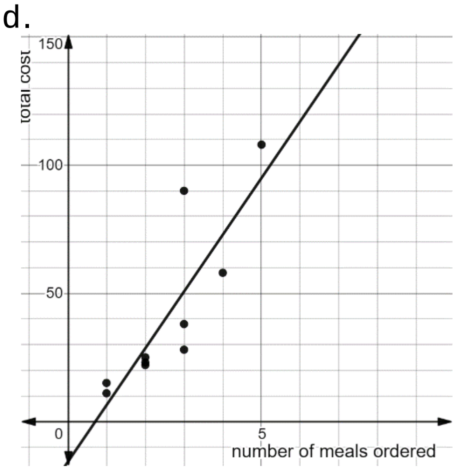
Algebra 1 Name Answer Key ___________
Unit 3 Date___________________
Lesson H1 Practice Period__________________

A restaurant took a random sample of customers ordering takeout meals. The data from the sample is shown.

Number of Meals ordered	1	2	5	3	2
Total Cost	\$15	\$23	\$108	\$28	\$22
Number of Meals ordered	4	3	3	1	2
Total Cost	\$58	\$90	\$38	\$11	\$25

This data is represented by a scatter plot with the function $c(x) = 22.17x - 15.83$ fit to the data, where c represents the total cost of the order and x represents the number of meals ordered.



For questions 1 through 5, complete the tables to find the residual values for each number of meals ordered.

1.

Number of Meals ordered	Actual Cost	Predicted Cost	Residual Values
1	\$15	\$6.34	8.66
1	\$11	\$6.34	4.66

2.

Number of Meals ordered	Actual Cost	Predicted Cost	Residual Values
2	\$23	\$28.51	−5.51
2	\$22	\$28.51	−6.51
2	\$25	\$28.51	−3.51

3.

Number of Meals ordered	Actual Cost	Predicted Cost	Residual Values
3	$28	$50.68	−22.68
3	$90	$50.68	39.32
3	$38	$50.68	−12.68

4.

Number of Meals ordered	Actual Cost	Predicted Cost	Residual Values
4	$58	$72.85	−14.85

5.

Number of Meals ordered	Actual Cost	Predicted Cost	Residual Values
5	$108	$95.02	12.98

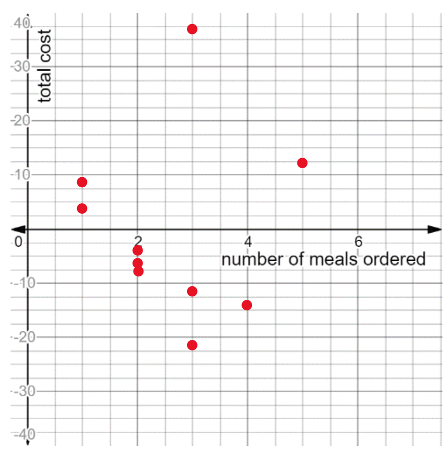
6. How many positive residuals does the data set have?

   4

7. What is the residual with the largest absolute value?

   39.32

8. Plot the residuals on the coordinate grid.



9. Explain whether the linear function is a good fit for this data set.

   While the points are randomly scattered, the number of points above the x-axis is not equal to the number of points below the x-axis, so a linear function would not be the best fit.